Project Design Phase- II
Solution Requirements (Functional & Non-functional)

Project Name	UCab – Cab Booking Web Application

FunctionalRequirements:

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story/ Sub-Task)
FR-1	User Registration	Registration through form, Gmail, or LinkedIn
FR-2	User Confirmation	Confirmation via email verification and OTP verification
FR-3	User Login & Authentication	Secure login using email/password and JWT token authentication
FR-4	Browse Available Cabs	View and filter cabs by type, seats, availability
FR-5	Cab Booking	Enter pickup, destination, date, time and confirm ride
FR-6	Fare Estimation	System calculates estimated fare based on distance and price per km
FR-7	Booking Management	Users can view booking history and cancel bookings
FR-8	Admin Dashboard	Admin can view statistics such as bookings, revenue, users
FR-9	Cab Management	Admin can add, edit, delete cabs and upload images
FR-10	User Management	Admin can view, update, activate/deactivate users
FR-11	Booking Management (Admin)	Admin can view and update booking status
FR-12	Image Upload	Admin can upload cab images using file upload system

Non-functionalRequirements:

Followingarethenon-functionalrequirementsoftheproposedsolution.

NFR No	Requirement	Description
NFR-1	Usability	The system should provide a simple and user-friendly interface allowing users to easily book rides.
NFR-2	Security	The system must use JWT authentication and encrypted passwords using bcrypt.
NFR-3	Reliability	The system should provide consistent and accurate service without data loss.
NFR-4	Performance	Pages and APIs should respond quickly even with multiple concurrent users.
NFR-5	Availability	The system should be accessible 24/7 with minimal downtime.
NFR-6	Scalability	The system should support increasing users and bookings without performance degradation.